Rohan Singh

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Education

University of Virginia Charlottesville, VA

BS in Computer Science – Minor in Data Science *December 2026*

- Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Computer Systems & Organization

Experience

Expedia Group Jun 2026 – Aug 2026

Machine Learning Engineer Intern | Python, FastAPI, Trino SQL *San Jose, CA*

- Prototyped a **FastAPI** service for OneKey, a **100M+ member rewards program**, with live profiles and safe fallbacks.
- Measured the program's first **randomized** causal readout on a live holdout: **+6.6%** relative booking lift at **p<0.001**.
- Showed the deployed targeting rule ranked below proportional random (**Qini -0.15**) and recommended against launch.
- Submitted a first-authored **CODE@MIT 2026** abstract on counterfactual logging and the limits of uplift targeting.
- Designed a typed reward API adopted by **2 sibling services**, with a fail-closed default when the model abstains.

Freddie Mac May 2025 – Jul 2025

Software Engineer Intern | Java, TypeScript, Spring Boot, Angular *McLean, VA*

- Implemented **Java Spring Boot** services to validate 120k accounting events and deployed them through **Jenkins CI/CD**.
- Designed **GraphQL** schema and resolvers spanning 200+ REST endpoints, cutting Angular client payload size **60%**.
- Benchmarked Playwright against Cypress, cutting test time **4x from 8 to 2 minutes** and driving the team's migration.
- Automated nightly UAT refreshes with Python and Jenkins, cutting load time from **2 hours to 10 minutes per run**.

Biocomplexity Institute Nov 2024 – May 2025

AI/ML Research Assistant | Python, Slurm *Charlottesville, VA*

- Developed Python graph-analysis pipelines with **cuGraph**, SciPy, and NetworkX over 1.2M-edge financial networks.
- Raised AUC-ROC from 0.55 to **0.76**, surfacing **250 anomalies at 89% precision** for downstream investigator review.
- Scaled end-to-end GPU anomaly detection with **cuGraph** to **2M+ nodes and edges** on UVA's Slurm cluster.

Cubic Corporation Jun 2024 – Aug 2024

Software Engineer Intern | JavaScript, React, Terraform *Sterling, VA*

- Launched an AWS-hosted React/Docusaurus developer portal on Terraform-managed infra, serving **500+ engineers**.
- Redesigned site flow and navigation after user tests; onboarding tickets fell **85%, from 90 to 13**, the next month.
- Added Algolia DocSearch and simplified navigation, reaching **95% top-3 accuracy at 120ms median query latency**.

Projects

Shortlist | Go, Python, PyTorch, Next.js

- Served exact retrieval over **89,585 movies** in Go with float16 vectors and history filtering at **4.0ms local median**.
- Reached **0.331 Recall@100 vs. 0.228 popularity** on **7,060** offline test users, covering **14.5%** of the evaluated catalog.
- Trained a **two-tower retriever** on a **32M**-rating catalog, aligning its objective to the 1–5 favorite onboarding flow.
- Shipped a Next.js app on Vercel with multi-title onboarding, natural reasons, saved lists, and **Playwright/Axe CI**.

Marathon Prep Bot | TypeScript, Node.js, Garmin FIT SDK

- Built a strict TypeScript pipeline decoding binary Garmin FIT files into **1,400+ CRC-validated activities** locally.
- Implemented Banister TRIMP, VDOT, and aerobic decoupling over per-second heart-rate streams to guide load planning.
- Automated 6-hour syncs with **550+ regression tests across 51 files**, keeping raw GPS and health data local.
- Shipped a live dashboard turning 44 months of activity data into weekly plans, training-load trends, and daily guidance.

Today (Marathon Prep Bot Companion) | Swift, SwiftUI, HealthKit, WorkoutKit

- Built the SwiftUI/Watch companion that executes coach plans and maps logged lifts to a **250+** exercise catalog.
- Closed the loop with offline-first **AES-256-GCM** sync; the server stores only ciphertext and public-safe aggregates.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, Go, Swift, SQL
Frameworks: Spring Boot, Node.js, Express, React, Next.js, FastAPI, GraphQL, PyTorch
Infrastructure & Data: AWS, GCP, Docker, Terraform, Jenkins, GitHub Actions, PostgreSQL, MySQL, MongoDB